ECP5 Stream Filter

Release 1.0.0

HDL Team

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Design

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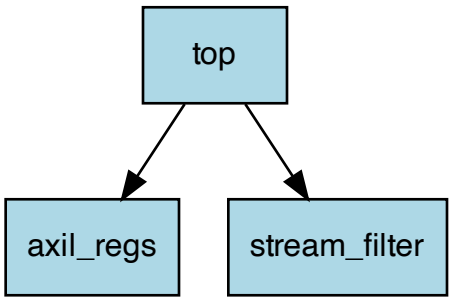
Chapter 1

ECP5 Stream Filter — Overview

This project contains 3 VHDL/SV modules.

Top-level module: *top*

1.1 Instantiation Tree



1.2 Modules

Module	Source File	Description	Instantiated by	Instantiates
top	`top.sv`	ECP5 design: an AXI4-Stream threshold filter with an AXI4-Lite register file. Bytes arriving on s_axis are forwarded to m_axis if they are at or above a	*(top)*	`axil_regs`, `stream_filter`
axil_re	`axil_reg sv`	AXI4-Lite register file for the stream filter. Four 32-bit registers: control (enable, clear), a read-only forwarded-byte	`top`	—
*stream_*	`stream_f sv`	Threshold filter for a byte stream. Forwards each byte that is at or above the threshold and drops the rest.	`top`	—

Chapter 2

top

2.1 top — Entity

Source file: `src/top.sv`.

ECP5 design: an AXI4-Stream threshold filter with an AXI4-Lite register file.

Bytes arriving on s_axis are forwarded to m_axis if they are at or above a threshold the bus master sets through the registers.

2.1.1 Annotated Source

Listing 1: src/top.sv

```
// ECP5 design: an AXI4-Stream threshold filter with an AXI4-Lite register file.
//
// Bytes arriving on s_axis are forwarded to m_axis if they are at or above a
// threshold the bus master sets through the registers.
module top (
    // System clock.
    input  logic   clk,
    // Synchronous active-high reset.
    input  logic   rst,
    // Register bus: control, status, threshold and scratch registers.
    axil_if.slave  s_axil,
    // Incoming byte stream.
    axis_if.slave  s_axis,
    // Filtered byte stream.
    axis_if.master m_axis
);
    logic        enable, clear;
    logic [7:0]  threshold;
    logic [15:0] count;

    axil_regs u_regs (
        .clk(clk), .rst(rst), .s_axil(s_axil),
        .enable(enable), .clear(clear), .threshold(threshold), .count(count)
    );

```

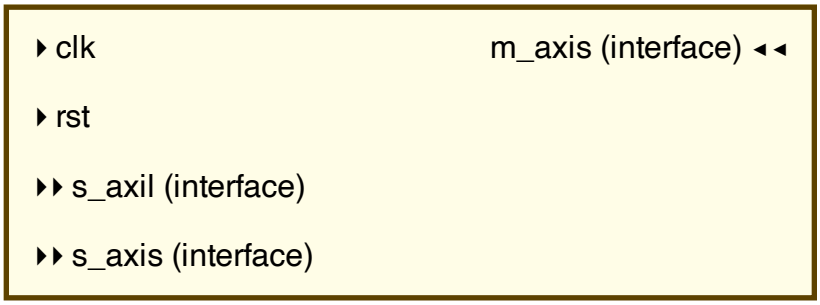
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```
    stream_filter u_filter (
        .clk(clk), .rst(rst), .enable(enable), .clear(clear), .threshold(threshold),
        .s_axis(s_axis), .m_axis(m_axis), .count(count)
    );
endmodule
```

2.2 top — Block Diagram

Auto-extracted from `top.sv`.



2.2.1 Ports

Port	Direction	Type	Description
`clk`	`in`	`logic`	System clock.
`rst`	`in`	`logic`	Synchronous active-high reset.
`s_axil`	`slave`	`axil_if`	Register bus: control, status, threshold and scratch registers.
`s_axis`	`slave`	`axis_if`	Incoming byte stream.
`m_axis`	`master`	`axis_if`	Filtered byte stream.

2.2.2 Interfaces

`axil_if` — used by `s_axil`.

Signal	Type	slave	Description
awaddr	logic[7:0]	in	Write address.
awvalid	logic	in	Write address valid.
awready	logic	out	Write address accepted.
wdata	logic[31:0]	in	Write data.
wstrb	logic[3:0]	in	Write byte strobes.
wvalid	logic	in	Write data valid.
wready	logic	out	Write data accepted.
bresp	logic[1:0]	out	Write response (OKAY = 0).
bvalid	logic	out	Write response valid.
bready	logic	in	Write response accepted.
araddr	logic[7:0]	in	Read address.
arvalid	logic	in	Read address valid.
arready	logic	out	Read address accepted.
rdata	logic[31:0]	out	Read data.
rresp	logic[1:0]	out	Read response (OKAY = 0).
rvalid	logic	out	Read data valid.
rready	logic	in	Read data accepted.

axis_if — used by s_axis, m_axis.

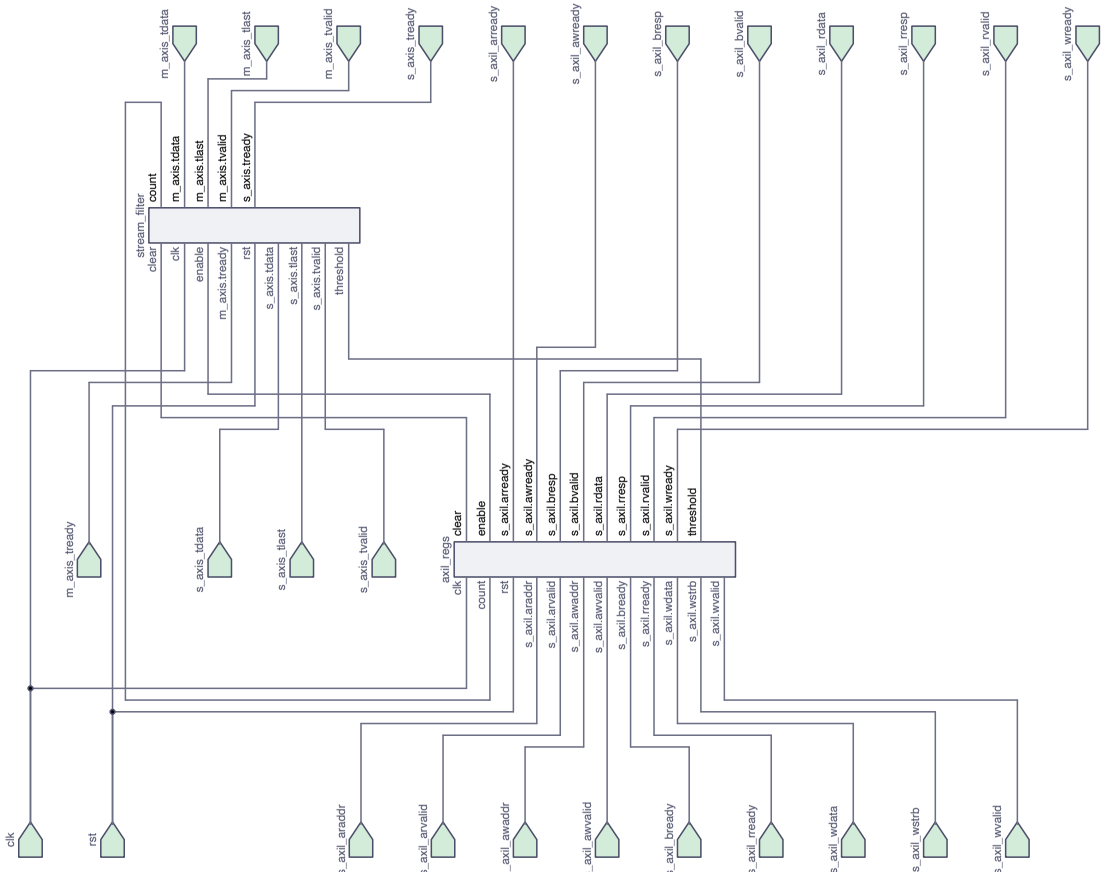
Signal	Type	slave	master	Description
tdata	logic[7:0]	in	out	Payload byte.
tvalid	logic	in	out	The source has a byte this cycle.
tready	logic	out	in	The sink accepts a byte this cycle.
tlast	logic	in	out	Last byte of a packet.

2.2.3 Signals

Signal	Type	Description
enable	logic	—
clear	logic	—
threshold	logic[7:0]	—
count	logic[15:0]	—

2.2.4 RTL Detail

Gate-level netlist generated by yosys and rendered with netlistsvg.


m_axis_tdata
m_axis_tlast
m_axis_tvalid
s_axis_tready
s_axil_arready
s_axil_awready
s_axil_bresp
s_axil_bvalid
s_axil_rdata
s_axil_rresp
s_axil_rvalid
s_axil_wready
stream_filter
count
m_axis.tdata
m_axis.tlast
m_axis.tvalid
s_axis.tready
clear
clk
enable
m_axis.tready
rst
s_axis.tdata
s_axis.tlast
s_axis.tvalid
threshold
m_axis_tready
s_axis_tdata
s_axis_tlast
s_axis_tvalid
axil_regs
clear
enable
s_axil.arready
s_axil.awready
s_axil.bresp
s_axil.bvalid
s_axil.rdata
s_axil.rresp
s_axil.rvalid
s_axil.wready
threshold
clk
count
rst
s_axil.araddr
s_axil.arvalid
s_axil.awaddr
s_axil.awvalid
s_axil.bready
s_axil.rready
s_axil.wdata
s_axil.wstrb
s_axil.wvalid
clk
rst
s_axil_araddr
s_axil_arvalid
s_axil_awaddr
s_axil_awvalid
s_axil_bready
s_axil_rready
s_axil_wdata
s_axil_wstrb
s_axil_wvalid

2.3 top — State Machine

No FSM detected in `top.sv`.

This module is either a structural wrapper (no combinational state logic) or uses a coding style not yet supported by the FSM extractor.

2.4 top — Timing Diagrams

All timing diagrams extracted from `top.sv`. Diagrams are sourced from `.. wavedrom::` blocks in the VHDL source comments above each process.

No wavedrom diagrams found in process comments.

Add `.. wavedrom::` blocks in the VHDL source above each process to have them appear here automatically.

2.5 top — Clock Domain Analysis

Auto-extracted from `top.sv`.

Note

No clocked processes detected. This module appears to be purely combinational.

2.6 top — Reset Domain Analysis

Auto-extracted from `top.sv`.

Note

No clocked processes detected. This module appears to be purely combinational.

2.7 Submodules

2.7.1 axil_regs

2.7.1.1 axil_regs — Entity

Source file: `src/axil_regs.sv`.

AXI4-Lite register file for the stream filter.

Four 32-bit registers: control (enable, clear), a read-only forwarded-byte count, the filter threshold, and a scratch register. Write strobes are ignored: every write updates the whole register. See registers/regs_axil_regs.toml.

Annotated Source

Listing 2: src/axil_regs.sv

```
1 // AXI4-Lite register file for the stream filter.
2 //
```

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```
// Four 32-bit registers: control (enable, clear), a read-only forwarded-byte
// count, the filter threshold, and a scratch register. Write strobes are
// ignored: every write updates the whole register. See registers/regs_axil_regs.toml.
module axil_regs (
    // System clock.
    input  logic        clk,
    // Synchronous active-high reset.
    input  logic        rst,
    // Register access from the bus master.
    axil_if.slave       s_axil,
    // Enables the stream filter.
    output logic        enable,
    // Holds the forwarded-byte counter at zero while high.
    output logic        clear,
    // Smallest byte value the filter forwards.
    output logic [7:0]  threshold,
    // Bytes forwarded since the counter was last cleared.
    input  logic [15:0] count
);
    logic [31:0] ctrl_r, threshold_r, scratch_r, rdata_r;
    logic        bvalid_r, rvalid_r;

    // Interface members are copied to plain signals first: Yosys cannot take a bit-select
    // of an interface member inside a submodule.
    wire [7:0]  awaddr = s_axil.awaddr;
    wire [7:0]  araddr = s_axil.araddr;
    wire [31:0] wdata  = s_axil.wdata;

    // A write is taken when both address and data are present and no response is pending.
    wire wr_fire = s_axil.awvalid && s_axil.wvalid && !bvalid_r;
    wire rd_fire = s_axil.arvalid && !rvalid_r;

    always_ff @(posedge clk) begin
        if (rst) begin
            ctrl_r      <= 32'd0;
            threshold_r <= 32'd0;
            scratch_r   <= 32'd0;
            rdata_r     <= 32'd0;
            bvalid_r    <= 1'b0;
            rvalid_r    <= 1'b0;
        end else begin
            if (wr_fire) begin
                bvalid_r <= 1'b1;
                case (awaddr[3:2])
                    2'd0:    ctrl_r      <= wdata;
                    2'd2:    threshold_r <= wdata;
                    2'd3:    scratch_r   <= wdata;
                    default: ;
                endcase
            end else if (s_axil.bready) begin
```

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```
                bvalid_r <= 1'b0;
            end

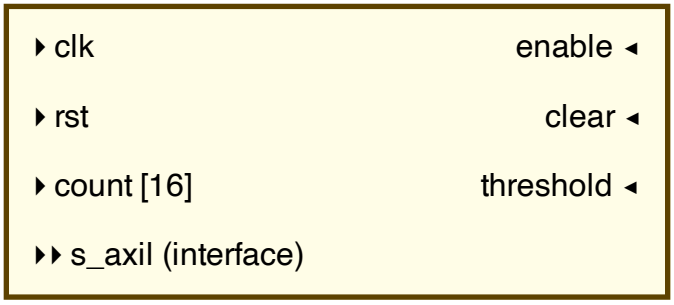
            if (rd_fire) begin
                rvalid_r <= 1'b1;
                case (araddr[3:2])
                    2'd0:    rdata_r <= ctrl_r;
                    2'd1:    rdata_r <= {16'd0, count};
                    2'd2:    rdata_r <= threshold_r;
                    default: rdata_r <= scratch_r;
                endcase
            end else if (s_axil.rready) begin
                rvalid_r <= 1'b0;
            end
        end
    end

    assign s_axil.awready = wr_fire;
    assign s_axil.wready  = wr_fire;
    assign s_axil.bresp   = 2'b00;
    assign s_axil.bvalid  = bvalid_r;
    assign s_axil.arready = rd_fire;
    assign s_axil.rdata   = rdata_r;
    assign s_axil.rresp   = 2'b00;
    assign s_axil.rvalid  = rvalid_r;

    assign enable    = ctrl_r[0];
    assign clear     = ctrl_r[1];
    assign threshold = threshold_r[7:0];
endmodule
```

2.7.1.2 axil_regs — Block Diagram

Auto-extracted from `axil_regs.sv`.



Ports

Port	Direction	Type	Description
clk	in	logic	System clock.
rst	in	logic	Synchronous active-high reset.
s_axil	slave	axil_if	Register access from the bus master.
enable	out	logic	Enables the stream filter.
clear	out	logic	Holds the forwarded-byte counter at zero while high.
threshold	out	logic[7:0]	Smallest byte value the filter forwards.
count	in	logic[15:0]	Bytes forwarded since the counter was last cleared.

Interfaces

axil_if — used by s_axil.

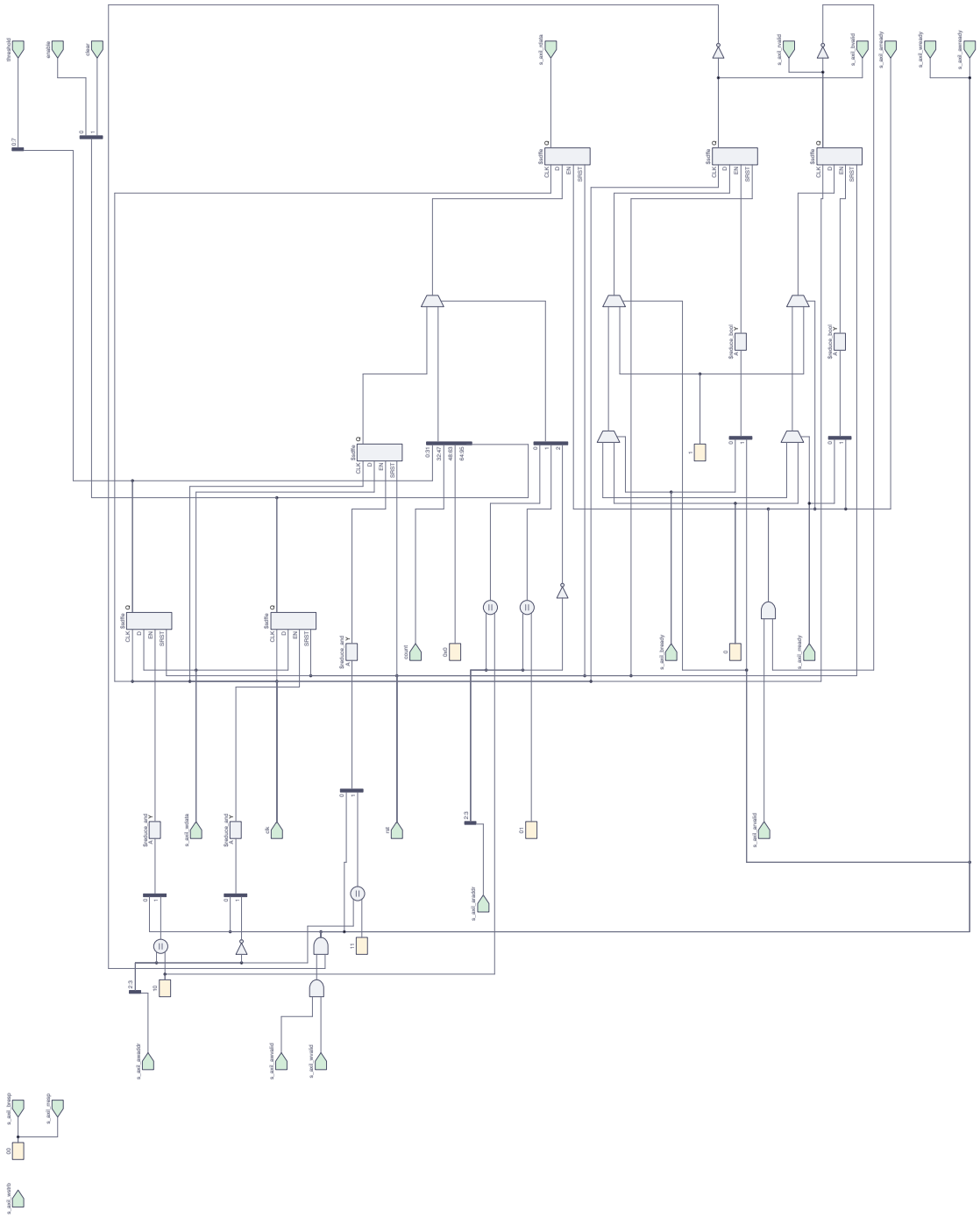
Signal	Type	slave	Description
awaddr	logic[7:0]	in	Write address.
awvalid	logic	in	Write address valid.
awready	logic	out	Write address accepted.
wdata	logic[31:0]	in	Write data.
wstrb	logic[3:0]	in	Write byte strobes.
wvalid	logic	in	Write data valid.
wready	logic	out	Write data accepted.
bresp	logic[1:0]	out	Write response (OKAY = 0).
bvalid	logic	out	Write response valid.
bready	logic	in	Write response accepted.
araddr	logic[7:0]	in	Read address.
arvalid	logic	in	Read address valid.
arready	logic	out	Read address accepted.
rdata	logic[31:0]	out	Read data.
rresp	logic[1:0]	out	Read response (OKAY = 0).
rvalid	logic	out	Read data valid.
rready	logic	in	Read data accepted.

Signals

Signal	Type	Description
ctrl_r	logic[31:0]	—
threshold_r	logic[31:0]	—
scratch_r	logic[31:0]	—
rdata_r	logic[31:0]	—
bvalid_r	logic	—
rvalid_r	logic	—

RTL Detail

Gate-level netlist generated by `yosys` and rendered with `netlistsvg`.



2.7.1.3 axil_regs — State Machine

No FSM detected in `axil_regs.sv`.

This module is either a structural wrapper (no combinational state logic) or uses a coding style not yet supported by the FSM extractor.

2.7.1.4 axil_regs — Timing Diagrams

All timing diagrams extracted from `axil_regs.sv`. Diagrams are sourced from `.. wavedrom::` blocks in the VHDL source comments above each process.

No wavedrom diagrams found in process comments.

Add `.. wavedrom::` blocks in the VHDL source above each process to have them appear here automatically.

2.7.1.5 axil_regs — Clock Domain Analysis

Auto-extracted from `axil_regs.sv`.

Note

No clocked processes detected. This module appears to be purely combinational.

2.7.1.6 axil_regs — Reset Domain Analysis

Auto-extracted from `axil_regs.sv`.

Note

All clocked processes use a single reset domain (`rst`, sync). No reset domain crossings detected.

Reset Domain

Process	Clock	Reset	Style
2	`clk`	`rst`	Sync

2.7.1.7 axil_regs — Register Map

Register	Offset	Access	Description
ctrl	0x00	r/w	Control register for the stream filter.
status	0x04	r	Filter status.
threshold	0x08	r/w	Filter threshold.
scratch	0x0C	r/w	General-purpose scratch register. Reads back what was written; no effect on the design.

ctrl (0x00)

Field	Type	Default	Description
`enable`	bit	0	Assert to let the filter forward bytes. While low every byte is dropped.
`clear`	bit	0	Hold high to hold the forwarded-byte counter at zero.

status (0x04)

Field	Type	Default	Description
`forwarded`	bit_vector(16)		Bytes forwarded since the counter was last cleared.

threshold (0x08)

Field	Type	Default	Description
`value`	bit_vector(8)	00000000	Smallest byte value that is forwarded. Bytes below it are dropped.

scratch (0x0C)

Field	Type	Default	Description
`value`	bit_vector(32)	00000000000000000(	Software scratch value.

2.7.2 stream_filter

2.7.2.1 stream_filter — Entity

Source file: `src/stream_filter.sv`.

Threshold filter for a byte stream.

Forwards each byte that is at or above the threshold and drops the rest. One register stage; counts the bytes it forwards.

Annotated Source

Listing 3: src/stream_filter.sv

```
1  // Threshold filter for a byte stream.
2  //
3  // Forwards each byte that is at or above the threshold and drops the rest.
4  // One register stage; counts the bytes it forwards.
5  module stream_filter (
6      // System clock.
7      input  logic        clk,
8      // Synchronous active-high reset.
9      input  logic        rst,
```

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```
    // Filtering is on while high; while low every byte is dropped.
    input  logic        enable,
    // Holds the counter at zero while high.
    input  logic        clear,
    // Smallest byte value that is forwarded.
    input  logic [7:0]  threshold,
    // Incoming bytes.
    axis_if.slave       s_axis,
    // Forwarded bytes.
    axis_if.master      m_axis,
    // Bytes forwarded since the counter was last cleared.
    output logic [15:0] count
);
    logic [7:0] tdata_r;
    logic       tvalid_r, tlast_r;

    // The output register is free when empty or being read this cycle.
    wire  can_load = !tvalid_r || m_axis.tready;
    wire  pass     = s_axis.tvalid && enable && (s_axis.tdata >= threshold);

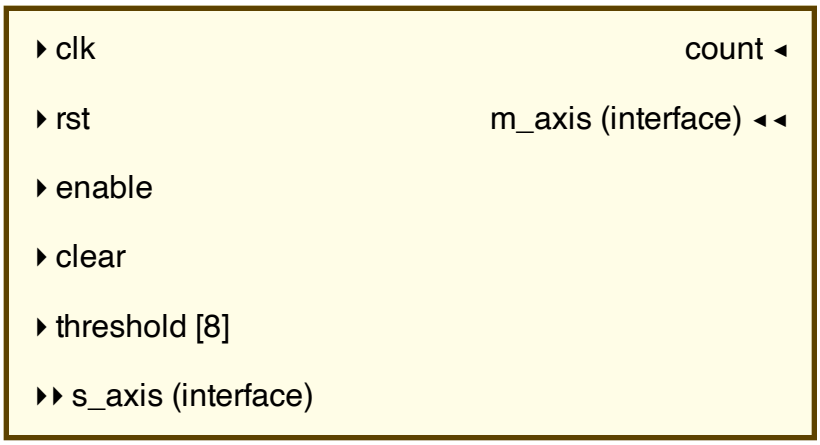
    always_ff @(posedge clk) begin
        if (rst || clear) begin
            count <= 16'd0;
        end else if (can_load && pass) begin
            count <= count + 16'd1;
        end

        if (rst) begin
            tvalid_r <= 1'b0;
        end else if (can_load) begin
            tvalid_r <= pass;
            tdata_r  <= s_axis.tdata;
            tlast_r  <= s_axis.tlast;
        end
    end

    assign s_axis.tready = can_load;
    assign m_axis.tdata  = tdata_r;
    assign m_axis.tvalid = tvalid_r;
    assign m_axis.tlast  = tlast_r;
endmodule
```

2.7.2.2 stream_filter — Block Diagram

Auto-extracted from `stream_filter.sv`.



Ports

Port	Direction	Type	Description
`clk`	`in`	`logic`	System clock.
`rst`	`in`	`logic`	Synchronous active-high reset.
`enable`	`in`	`logic`	Filtering is on while high; while low every byte is dropped.
`clear`	`in`	`logic`	Holds the counter at zero while high.
`threshold`	`in`	`logic[7:0]`	Smallest byte value that is forwarded.
`s_axis`	`slave`	`axis_if`	Incoming bytes.
`m_axis`	`master`	`axis_if`	Forwarded bytes.
`count`	`out`	`logic[15:0]`	Bytes forwarded since the counter was last cleared.

Interfaces

`axis_if` — used by `s_axis`, `m_axis`.

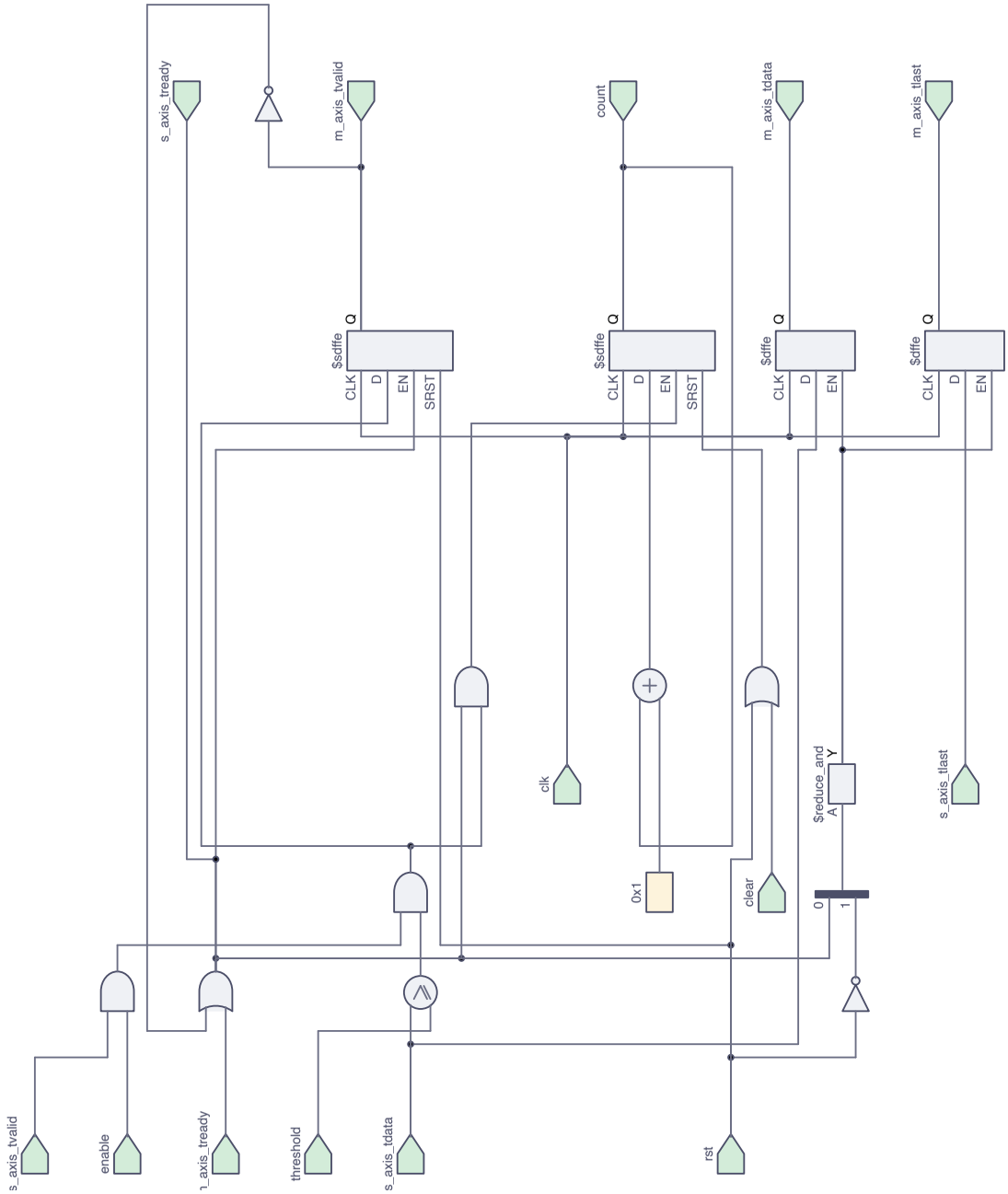
Signal	Type	slave	master	Description
`tdata`	`logic[7:0]`	in	out	Payload byte.
`tvalid`	`logic`	in	out	The source has a byte this cycle.
`tready`	`logic`	out	in	The sink accepts a byte this cycle.
`tlast`	`logic`	in	out	Last byte of a packet.

Signals

Signal	Type	Description
`tdata_r`	`logic[7:0]`	—
`tvalid_r`	`logic`	—
`tlast_r`	`logic`	—

RTL Detail

Gate-level netlist generated by `yosys` and rendered with `netlistsvg`.


s_axis_tready
m_axis_tvalid
count
m_axis_tdata
m_axis_tlast
$sdffe
$sdffe
$dffe
$dffe
Q
CLK
D
EN
SRST
clk
0x1
clear
$reduce_and
A
Y
0
1
s_axis_tlast
s_axis_tvalid
enable
m_axis_tready
threshold
s_axis_tdata
rst

2.7.2.3 stream_filter — State Machine

No FSM detected in `stream_filter.sv`.

This module is either a structural wrapper (no combinational state logic) or uses a coding style not yet supported by the FSM extractor.

2.7.2.4 stream_filter — Timing Diagrams

All timing diagrams extracted from `stream_filter.sv`. Diagrams are sourced from `.. wavedrom::` blocks in the VHDL source comments above each process.

No wavedrom diagrams found in process comments.

Add `.. wavedrom::` blocks in the VHDL source above each process to have them appear here automatically.

2.7.2.5 stream_filter — Clock Domain Analysis

Auto-extracted from `stream_filter.sv`.

Note

No clocked processes detected. This module appears to be purely combinational.

2.7.2.6 stream_filter — Reset Domain Analysis

Auto-extracted from `stream_filter.sv`.

Note

No clocked processes detected. This module appears to be purely combinational.

Chapter 3

Register Maps

Chapter 4

Testbenches

Framework	Testbenches	Tests
VUnit	1	3
cocotb	1	3 (1 skipped)
Directed	1	2
Total	3	8 (1 skipped)

4.1 VUnit

4.1.1 tb_stream_filter

File
`tb/vunit/tb_stream_filter.sv`

Tests module
stream_filter

Purpose
Testbench for stream_filter. Drives bytes into s_axis and checks which ones come out of m_axis: the filter forwards bytes at or above the threshold and drops the rest.

Tests
3 tests

Test	Purpose
`passes_at_or_above_threshold`	A byte equal to or above the threshold appears on m_axis.
`drops_below_threshold`	A byte below the threshold is dropped.
`counts_forwarded_bytes`	The counter counts only the bytes that were forwarded.

4.2 cocotb

4.2.1 test_axil_regs

File
`tb/cocotb/test_axil_regs.py`

Tests module
axil_regs

Purpose
Register file tests for axil_regs.

Tests
3 tests, 1 skipped

Test	Purpose
`reset_values`	Every register reads back zero after reset.
`scratch_write_read`	A value written to the scratch register reads back unchanged.
`threshold_drives_filter` *(skipped)*	The threshold register changes which bytes the filter forwards (needs the full top level).

4.3 Directed

4.3.1 tb_top

File
`tb/directed/tb_top.sv`

Tests module
top

Purpose
Directed testbench for the top level. Programs the filter through the AXI4-Lite bus, then streams bytes through it, with no verification framework: each scenario below is marked with a TEST comment.

Tests
2 tests

Test	Purpose
`register_write_read`	a value written to scratch reads back unchanged
`filter_pass_through`	bytes at or above the threshold reach m_axis, others do not

Chapter 5

Build

5.1 Summary

Yosys	0.56+186
Device	ECP5 25k class
Reports generated	2026-10-31 22:35
Documented at	v3.12.0-26-g3652455

5.2 Utilization

5.2.1 Device utilisation (nextpnr)

Resource	Used	Available	%
TRELLIS_IO	122	197	61
DCCA	1	56	1
TRELLIS_FF	156	24,288	0
TRELLIS_COMB	147	24,288	0

24 other resource types unused.

5.2.2 By module (Yosys)

Module	LUTs	FFs	BRAMs	DSPs
top	113	156	0	0

5.3 Timing — nextpnr

Clock	Fmax (MHz)	Target (MHz)	Status
clk	220.41	100.0	✓ PASS

5.4 Warnings

146 warnings in the build logs; the most frequent:

Message	Count
IO '...' is unconstrained in LPF and will be automatically placed	121
Identifier '...' is implicitly declared.	24
The network is combinational.	1

Chapter 6

Indices

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